

Prepare Your Application: Before you begin testing, you need to ensure that your application has been optimized for foldable screens. You can use tools like Jetpack WindowManager to help you manage window layouts and display features.

Identify Key Scenarios to Test: Identify the critical functionalities and user flows of your app that will be affected by the foldable form factor. These can include transitioning from the cover display to the main display, running multiple apps simultaneously, or using the device in different postures.

Run Your Tests: Conduct your tests according to the types of tests you've chosen (discussed below). Make sure to document any issues or anomalies that arise during testing.

Review and Interpret Your Results: After conducting your tests, review the data collected. Look for patterns, recurring issues, and areas where the app performed well.

Different types of tests to conduct

Functionality Testing: This type of testing checks if your app functions as expected on foldable screens. It includes testing the app in different configurations (folded, unfolded, multi-window mode), checking how it transitions between screens, and ensuring it correctly detects the state of the device.



Usability Testing: Usability testing assesses how user-friendly your app is on a foldable screen. It involves testing the interface and navigation in different device postures and screen configurations. The goal is to ensure that the user has a smooth, intuitive experience regardless of how they're using the device.

Performance Testing: Performance testing evaluates how well your app runs on a foldable device. This can involve testing the speed of transitions, the responsiveness of the app in different configurations, and how well it handles running multiple apps at once.

Compatibility Testing: This type of testing ensures that your app works seamlessly with the underlying operating system and other apps. It includes testing how your app interacts with system features, such as multi-window mode, as well as how it operates when other apps are running.

Resilience Testing: Foldable devices can be used in a variety of postures and configurations, which can potentially strain your app. Resilience testing involves putting your app through these rigorous use cases to ensure it remains stable and performs well.

Functionality Testing

Functionality testing is the first step. This ensures that all features of your application work correctly on a foldable device.

Test the app in both folded and unfolded states. The app should work seamlessly in both situations.

Check the App Continuity feature. The transition from the cover display to the main display should be smooth, and the user should be able to continue their task without interruption.

Test Multi-Active Window functionality. Your app should be able to run simultaneously with other apps, and interaction between these apps should be flawless.

Test the app in different orientations and postures. The content should adapt and display correctly.

Usability Testing

Usability testing focuses on the user's experience. It's crucial to ensure your app not only functions correctly but is also intuitive and user-friendly on foldable devices.

Check the user interface and ensure it adapts properly to different screen sizes and layouts.

Test navigation within the app. The navigation should be easy and intuitive in both folded and unfolded states.

Evaluate the text size and touch targets. They should be appropriately sized for both states.

Performance Testing

Performance testing ensures your app runs smoothly without draining the device's resources.

Test your app's performance in both folded and unfolded states. The performance should be consistent across different device states.

Check the app's response time, battery consumption, and memory usage. These should be within acceptable limits.

Compatibility Testing

Compatibility testing ensures your app works with all the features of the foldable device. Test your app with different system features of the device, like the camera, GPS, and sensors. Check your app's compatibility with the device's OS and its features.

How to interpret test results?

When interpreting the results of your tests, it's essential to understand what the data is telling you. For functionality tests, any failure indicates a bug that needs to be fixed. Usability tests may highlight areas of your app's user interface that are confusing or difficult to navigate on a foldable screen. Perfor-